

Rivertek RT-2050 Pressure Transmitter -- Technical Manual

PRODUCT OVERVIEW

The Rivertek RT-2050 is a high-precision pressure transmitter designed for reliable pressure monitoring in water treatment processes. Engineered with advanced sensor technology, the RT-2050 offers exceptional accuracy, stability, and durability, making it ideal for demanding environments within water and wastewater treatment plants. This device measures pressure ranges from 0 to 10 bar and features a robust housing, ensuring optimal performance even in harsh conditions.

Key features of the RT-2050 include:

- Digital signal processing for enhanced accuracy.
- Wide temperature operating range (-20°C to 80°C).
- IP68-rated enclosure for water and dust resistance.
- RS-485 communication protocol for easy integration into existing systems.
- Configurable output options for flexibility in various applications.

TECHNICAL SPECIFICATIONS

- Model: Rivertek RT-2050
- Pressure Range: 0 to 10 bar
- Accuracy: $\pm 0.5\%$ of Full Scale
- Output Signal: 4-20 mA or RS-485
- Power Supply: 24 VDC $\pm 10\%$
- Operating Temperature: -20°C to 80°C
- Storage Temperature: -40°C to 85°C
- Enclosure Rating: IP68
- Process Connection: 1/2" NPT
- Weight: 1.2 kg
- Certification: CE, ISO 9001

CALIBRATION PROCEDURE

To ensure optimal performance and accuracy of the Rivertek RT-2050, regular calibration is recommended. The following steps outline the calibration procedure:

- **Preparation****: Ensure that the transmitter is disconnected from the power supply. Gather necessary tools, including a calibrated pressure source, multimeter, and calibration software (if applicable).
- **Connect Calibration Equipment****: Attach the pressure source to the transmitter's process connection, ensuring a secure fit to prevent leaks. Connect the multimeter to the output terminals of the RT-2050.
- **Apply Pressure****: Gradually apply pressure to the transmitter using the calibrated pressure source. Start at 0 bar and incrementally increase to the full scale (10 bar).

4. **Record Output**: At each pressure increment (0, 2, 4, 6, 8, and 10 bar), record the corresponding output signal from the multimeter. Ensure that the output remains stable before recording the value.
5. **Adjust Calibration**: Compare the recorded output values against the expected output (4-20 mA). If discrepancies are found, adjust the calibration settings using the calibration software or the built-in adjustment mechanism as per the user manual.
6. **Final Check**: Once adjustments are made, repeat the pressure application to confirm that the output signal aligns with the applied pressure. Document all calibration results for future reference.
7. **Reconnect**: After successful calibration, disconnect the pressure source, remove the multimeter, and reconnect the transmitter to the power supply.

ERROR CODES

The Rivertek RT-2050 includes built-in diagnostics to assist with troubleshooting. Below are common error codes, their causes, and recommended resolutions:

- **Error Code 101: Sensor Failure**
 - Cause: Internal sensor malfunction.
 - Resolution: Power cycle the device. If the error persists, replace the transmitter.
- **Error Code 102: Overpressure**
 - Cause: Pressure exceeds maximum rated limit.
 - Resolution: Disconnect the transmitter from the process. Reduce pressure to within specified limits.
- **Error Code 103: Communication Error**
 - Cause: Faulty wiring or communication disruption.
 - Resolution: Check wiring connections and ensure the RS-485 settings match the system configuration.
- **Error Code 104: Calibration Error**
 - Cause: Calibration values exceed acceptable limits.
 - Resolution: Recalibrate the device following the calibration procedure outlined above.
- **Error Code 105: Supply Voltage Error**
 - Cause: Power supply voltage is outside the acceptable range.
 - Resolution: Verify and correct the power supply voltage to 24 VDC $\pm 10\%$.

MAINTENANCE

Regular maintenance of the Rivertek RT-2050 is essential for sustained performance and longevity. Follow these guidelines to maintain the transmitter:

1. **Visual Inspection**: Regularly inspect the transmitter for any signs of physical damage, corrosion, or leaks. Ensure that the housing is free from dirt and debris.
2. **Cleaning**: Clean the exterior of the transmitter using a soft cloth. Avoid using abrasive materials or solvents that may damage the enclosure.
3. **Functional Testing**: Conduct periodic functional tests to verify that the device is operating within

specified parameters. This should include checking the output signal against known pressure values.

4. ****Calibration Schedule****: Establish a calibration schedule based on usage and environmental conditions. Typically, calibration should occur at least once a year or as recommended by operational guidelines.

5. ****Replacement Parts****: Keep a stock of critical replacement parts, such as O-rings and seals, to ensure quick repairs when needed.

By adhering to these maintenance practices, the Rivertek RT-2050 will deliver reliable performance for years to come, ensuring efficient water treatment processes.